TECHNICAL DILIGENCE REVIEW - RECORD 024

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-024

1. Scope of the technical-debt review and its basis

The present report is one of a series of technical reports; this one takes the technical-debt register, end-of-life dependency exposure and remediation follow-through and reports against the evidence available by 2026-07-15. The subject of the present report is the technical-debt register, end-of-life dependency exposure and remediation follow-through, examined for Ardent Peak by Wrenfield Technical Advisory LLP over 3 weeks ending 2026-07-15. The reviewer found the interview evidence for the present report candid, and in two cases more candid than the written record. R. Ostrowski interviewed 13 engineers and managers for the report at TDR-2026-34; two declined to be quoted on one topic and that is noted where it matters. R. Ostrowski has resisted rounding the findings towards a comfortable conclusion in either direction.

Method in report TDR-2026-34 was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. R. Ostrowski led the work behind TDR-2026-34, combining document review, 13 interviews and hands-on inspection of running systems. Not every finding in the present report is accepted by management: 3 are declined, and the declinatures are reproduced rather than argued away. Where management has declined to act, this report preserves the decision and the stated rationale so a post-close plan can revisit it. R. Ostrowski has separated what the company has built from what the company can demonstrate about what it has built.

The convention throughout TDR-2026-34 is that an absence of evidence is reported as such rather than as evidence of absence. R. Ostrowski treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. 3 items in this report sit in a declined state, and the reviewer regards one of them as wrongly declined. 3 declined items are carried in TDR-2026-34, of which two lack any documented compensating control. R. Ostrowski has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

The principal limitation on the present report is time: 3 weeks was enough to test the control path and not enough to test the reporting path. One limitation of the report at TDR-2026-34 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. This report concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. The overall conclusion of the report at TDR-2026-34 is that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. Where this report uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

The population for this report was the service inventory as at 2026-07-15; 11 entries were examined and the rest were reviewed only at inventory level. 11 services were examined in detail for report TDR-2026-34, chosen for their position in the dependency map rather than at random. R. Ostrowski was granted read access to the code, the pipeline and the incident tracker for the purposes of TDR-2026-34; production access was not requested. The approach R. Ostrowski took in report TDR-2026-34 was to test the company's own assertions rather than to restate them. Nothing in this report is intended as a legal opinion or as advice on the adequacy of insurance cover.

Minutes of every the report at TDR-2026-34 interview were returned to the participant, and three came back with substantive corrections. The present report draws on 13 interviews, including the engineering owners of every component examined in detail. R. Ostrowski could not verify one class of claim in the present report, namely the behaviour of components deployed only at the two sites not visited. The reviewer notes that the report at TDR-2026-34 was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The technical-debt register, end-of-life dependency exposure and remediation follow-through was examined over 3 weeks, which sets a real limit on the depth reached in the later sections.

Schedule V - Costed technical-debt and migration register (TDR-2026-34)

Item	Area	Component	Effort estimate	Costed	Target date	Status
Partition the fleet allocator write path	scalability	Fleet Allocator	3.5 engineer-quarters	$420,000	2027-03-31	in design

Item	Area	Component	Effort estimate	Costed	Target date	Status
Replace the unmaintained scan-matching library in the localization fallback	IP and support	Localization Engine	1.5 engineer-quarters	$185,000	2026-12-31	committed
Retire the vendored geometry library in costmap fusion	support	Costmap Fusion	1 engineer-quarter	$128,000	2027-06-30	backlog
Raise robot software line coverage above sixty per cent on the perception path	quality	Costmap Fusion	2 engineer-quarters	$240,000	2027-03-31	in progress
Bring the second bench rig to the current sensor configuration	test capability	Hardware-in-the-Loop Rig	0.5 engineer-quarters	$96,000	2026-09-30	committed
Make the nightly simulation suite blocking	quality	Simulation Harness	1 engineer-quarter	$115,000	2026-11-30	blocked
Pin the full simulation build configuration and enable a determinism check	tooling	Simulation Harness	0.5 engineer-quarters	$68,000	2026-05-31	overdue
Replace the archived charting library in the operator console	security supportability	Remote Operations Console	1 engineer-quarter	$132,000	2027-01-31	backlog
Upgrade the streaming platform before its support window closes	support	Event Bus	1.5 engineer-quarters	$196,000	2027-06-30	committed
Migrate the deprecated warehouse transformation function	support	Analytics Warehouse Loader	0.5 engineer-quarters	$54,000	2026-12-31	committed
Re-converge the four forked site provisioning templates	configuration	Site Provisioning Service	0.5 engineer-quarters	$61,000	2026-09-30	in progress
Extend the edge buffer to twelve hours	resilience	Edge Gateway Agent	1 engineer-quarter	$144,000	2026-10-31	in progress
Replace the fallback scan matcher with the current implementation	performance	Localization Engine	1.5 engineer-quarters	$178,000	2026-12-31	in design
Introduce a second bench rig queue to cut firmware lead time	delivery	Hardware-in-the-Loop Rig	0.5 engineer-quarters	$88,000	2027-03-31	backlog
Add per-region credentials for the paging provider	security	Alarm Router	0.25 engineer-quarters	$34,000	2026-09-30	committed
Encrypt the on-robot localization map cache	security	Localization Engine	1 engineer-quarter	$126,000	2026-10-15	committed
Enforce a two-person rule on OTA artefact signing	security	OTA Updater	0.5 engineer-quarters	$72,000	2026-11-30	in design
Aggregate the	safety evidence	Safety Controller	0.75 engineer-	$104,000	2027-03-31	backlog

Item	Area	Component	Effort estimate	Costed	Target date	Status
safety controller fault log to the cloud continuously			quarters			
Measure closing rate for the separation monitor	safety	Speed and Separation Monitor	2 engineer-quarters	$268,000	2026-08-31	in progress
Validate and ship the wheel slip detector	safety	Wheel Odometry Filter	1.5 engineer-quarters	$162,000	2026-09-15	in progress
Refit the battery state-of-health estimator on opportunity-charge data	quality	Battery Management Service	1 engineer-quarter	$138,000	2026-09-30	in progress
Move the calibration record for the payload scale to the cloud	quality	Payload Scale Interface	0.25 engineer-quarters	$42,000	2027-01-31	backlog
Split the shared database instance used by the allocator and the charge scheduler	reliability	Charge Scheduler	0.75 engineer-quarters	$92,000	2026-12-31	committed
Publish the modified files carrying file-level copyleft obligations	IP hygiene	Costmap Fusion	0.25 engineer-quarters	$28,000	2026-09-30	committed

2. The costed register: TD-11 and the funded items

Two templates remain forked as at the review date is the point the reviewer would put to management on TD-11 on Site Provisioning Service. What Site Provisioning Service item TD-11 protects against is the fact that it template drift produced a go-live failure in May 2026. Programme item TD-11 is behind a platform change that has its own target date, which is why the target date is later than the effort estimate implies. register item TD-11 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

Safety Controller item TD-18 matters because it investigation evidence is lost between weekly extractions. TD-18 on Safety Controller cannot start earlier than it does because it is behind a platform change that has its own target date. Under safety evidence, entry TD-18 is to aggregate the safety controller fault log to the cloud continuously. register item TD-18 was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

Replace the archived charting library in the operator console is entered at TD-08 as an security supportability item. About TD-08 on Remote Operations Console it should be said that no advisory affects the current version, which is why it ranks low. Whether TD-08 happens depends on funding, and it is funded from the second quarter onward. security supportability item TD-08 is the clearest instance in TDR-2026-34 of engineering practice being better than the documentation suggests.

2027-01-31 is the target for TD-22, whose status is backlog. Ahead of two other items that depend on it is where programme item TD-22 sits in the programme. Reading TD-22 against the funded plan, a small item that keeps returning as a field support cost. TD-22 was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

For TD-24 the team estimates 0.25 engineer-quarters, which the reviewer converted to $28,000 for comparison. Progress on IP hygiene item TD-24 is recorded as committed against a 2026-09-30 date. Reading the programme plan, register item TD-24 is behind a customer maintenance window that is not yet agreed. TD-24 on Costmap Fusion was checked against the corresponding entry in the company's own register, and the register was current.

The estimate for Hardware-in-the-Loop Rig item TD-14 is 0.5 engineer-quarters and $88,000. Introduce a second bench rig queue to cut firmware lead time is entered at Hardware-in-the-Loop Rig item TD-14 as an delivery item. About

TD-14 it should be said that requires capital for a third rig rather than engineering effort. TD-14 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Last in a three-item chain, so its date is the least reliable on the register is where Fleet Allocator item TD-01 sits in the programme. The reviewer's comment on Fleet Allocator item TD-01 is that the load-test series identifies this as the single binding scalability constraint. TD-01 is funded from the second quarter onward in the engineering plan the reviewer was shown. scalability item TD-01 would be a candidate for the first post-close engineering review under any reasonable plan.

OTA Updater carries security item TD-17, the work to enforce a two-person rule on OTA artefact signing. On funding, programme item TD-17 is unfunded and dependent on capital rather than engineering effort. The reviewer's comment on TD-17 is that the control is designed but the signing tool does not enforce it. OTA Updater item TD-17 was reviewed against the position twelve months earlier, and nothing measurable has changed.

The plan treats support item TD-03 as partly funded, with the second phase unbudgeted, which the reviewer checked against the headcount model. Effort on register item TD-03 is estimated at 1 engineer-quarter, costed at $128,000. Costmap Fusion item TD-03 matters because it no upstream fixes for a component on the safety-adjacent path. Costmap Fusion item TD-03 is reported at the evidence strength available and has not been promoted to fit the narrative.

TD-07 covers the work to pin the full simulation build configuration and enable a determinism check. Sequencing of TD-07 on Simulation Harness matters: it is last in a three-item chain, so its date is the least reliable on the register. Funding for register item TD-07 is funded in the current financial year, and the register does not say so. register item TD-07 is treated as material to the investment case rather than as a routine engineering item.

Unfunded and dependent on capital rather than engineering effort is the funding position on programme item TD-12. Effort on register item TD-12 is estimated at 1 engineer-quarter, costed at $144,000. About programme item TD-12 it should be said that the site alert landed first; the buffer extension needs a storage change on the compute module. programme item TD-12 is the entry that changed most between the first and second document requests behind TDR-2026-34.

Programme item TD-16 matters because it a stolen compute module discloses customer warehouse layouts. The reviewer's qualification on Localization Engine item TD-16 is that depends on the next compute module revision. The plan treats security item TD-16 as partly funded, with the second phase unbudgeted, which the reviewer checked against the headcount model. programme item TD-16 is the entry that changed most between the first and second document requests behind TDR-2026-34.

Schedule T - Third-party component and licence inventory (TDR-2026-34)

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
nav-graph-solver	2.4.1	Apache-2.0	no	linked into the planner as a static library	approved 2024-02-19	Motion Planner
geom2d-fusion	0.9.7	LGPL-2.1	weak	dynamically linked by the fusion pipeline	approved 2023-08-04	Costmap Fusion
scanmatch-legacy	1.2.0	GPL-2.0	strong	linked into the localization fallback path	not approved	Localization Engine
grpc-transport	1.62.0	Apache-2.0	no	service-to-service transport across the cloud	approved 2024-05-30	Event Bus
tsdb-client	3.1.4	MIT	no	telemetry writes from the ingest service	approved 2024-09-11	Telemetry Ingest
chartkit	4.8.2	BSD-3-Clause	no	operator console visualisations	approved 2023-11-27	Remote Operations Console
physics-core-fork	0.14.3	Zlib	no	simulation physics	approved with conditions 2023-04-18	Simulation Harness
rtos-shim	1.0.9	BSD-2-Clause	no	real-time helper layer on the robot	approved 2024-01-23	Motion Planner
protobuf-runtime	25.3	BSD-3-Clause	no	message encoding	approved 2024-	Config

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
				across edge and cloud	04-08	Distribution
sqlite-embedded	3.45.1	public domain dedication	no	on-robot local state	approved 2023-06-15	Edge Gateway Agent
opencv-subset	4.9.0	Apache-2.0	no	image preprocessing for grasp proposals	approved 2024-07-02	Gripper Controller
pointcloud-utils	2.1.6	MPL-2.0	weak	point cloud filtering in fusion	approved 2024-03-14	Costmap Fusion
yaml-parse	0.15.2	MIT	no	configuration parsing on the robot	approved 2023-09-29	Config Distribution
jwt-verify	5.2.1	MIT	no	token verification in the identity service	approved 2024-06-21	Identity and Authorization
kafka-client	3.6.1	Apache-2.0	no	event bus producer and consumer clients	approved 2024-02-27	Event Bus
media-transcode	6.0.0	LGPL-2.1 with an optional GPL build	weak	video snippet transcoding	approved with conditions 2024-08-16	Video Snippet Service
wms-adapter-sdk	2.2.0	commercial, per-seat	no	one customer's warehouse system adapter	approved 2023-12-05	WMS Connector
lidar-vendor-sdk	4.1.2	commercial, redistributable binary	no	lidar driver	approved 2023-05-22	Lidar Driver
safety-controller-toolchain	3.0.4	commercial, development use only	no	safety controller build toolchain	approved 2023-02-10	Safety Controller
cell-model-lib	1.8.0	Apache-2.0	no	battery state estimation	approved 2024-10-01	Battery Management Service
grid-render	3.3.5	GPL-3.0 with a commercial alternative	strong	map authoring geometry rendering	approved with conditions 2023-07-19	Map Authoring Studio
http-router	1.9.4	MIT	no	cloud service routing	approved 2024-01-09	Order Broker
otel-instrument	1.24.0	Apache-2.0	no	tracing across cloud services	approved 2024-05-14	Fleet Health Monitor
pgbouncer-managed	1.21.0	ISC	no	database connection pooling	approved 2024-03-05	Fleet Allocator
wheel-slip-model	0.3.1	AGPL-3.0	strong	prototype slip detector, not shipped	not approved	Wheel Odometry Filter
csv-export	2.0.3	MIT	no	billing and usage exports	approved 2023-10-17	Billing Usage Meter
pager-provider-sdk	3.4.0	commercial, subscription	no	alarm dispatch	approved 2023-03-28	Alarm Router
k8s-operator-framework	1.31.0	Apache-2.0	no	cloud deployment controllers	approved 2024-09-24	Site Provisioning Service
image-signing	2.5.0	Apache-2.0	no	artefact signature verification	approved 2025-01-15	OTA Updater
warehouse-sim-assets	1.1.0	CC BY-SA 4.0	strong	simulation scenario assets	approved with conditions 2024-11-06	Simulation Harness

3. End-of-life dependencies and supportability

TD-02 on Localization Engine matters because it strong-copyleft exposure in shipped robot software plus an unsupported dependency. Under IP and support, entry TD-02 on Localization Engine is to replace the unmaintained scan-matching library in the localization fallback. 2026-12-31 is the target for IP and support item TD-02, whose status is committed. IP and support item TD-02 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-34.

TD-15 on Alarm Router is committed with a target date of 2026-09-30. Raised in the 2026 assessment and accepted for remediation is the point the reviewer would put to management on security item TD-15. Programme item TD-15 is unfunded and dependent on capital rather than engineering effort in the engineering plan the reviewer was shown. TD-15 on Alarm Router illustrates why TDR-2026-34 separates the severity of a finding from the confidence behind it.

Upgrade the streaming platform before its support window closes is entered at Event Bus item TD-09 as an support item. Support item TD-09 appears in the register and is funded and already staffed. Reading TD-09 on Event Bus against the funded plan, the upgrade requires a maintenance window at every site. programme item TD-09 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The business consequence recorded against Localization Engine item TD-13 is that it relocalisation exceeds the operator prompt timeout at sparse sites. TD-13 is sequenced behind a customer maintenance window that is not yet agreed. Programme item TD-13 is not straightforward: carries the same dependency as the copyleft removal. programme item TD-13 illustrates why TDR-2026-34 separates the severity of a finding from the confidence behind it.

Register item TD-05 is committed with a target date of 2026-09-30. For programme item TD-05 the team estimates 0.5 engineer-quarters, which the reviewer converted to $96,000 for comparison. On register item TD-05 the reviewer records that outstanding since the April 2025 lidar timing incident. TD-05 is the clearest instance in TDR-2026-34 of a control that exists on paper and is weaker in practice.

Left in place, the condition behind TD-04 on Costmap Fusion defect escape rate on the fusion path is the highest in the fleet. The reviewer's comment on TD-04 on Costmap Fusion is that coverage moved from 51 to 58 per cent over the reviewed period. Programme item TD-04 cannot start earlier than it does because it is behind a customer maintenance window that is not yet agreed. programme item TD-04 is reported at the evidence strength available and has not been promoted to fit the narrative.

Charge Scheduler carries TD-23, the work to split the shared database instance used by the allocator and the charge scheduler. Left in place, the condition behind TD-23 one noisy query affects task assignment. Status on Charge Scheduler item TD-23 is committed; the target is 2026-12-31. TD-23 on Charge Scheduler was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Progress on register item TD-10 is recorded as committed against a 2026-12-31 date. Item TD-10 on Analytics Warehouse Loader in the debt register is to migrate the deprecated warehouse transformation function, and it bears on Analytics Warehouse Loader. The dependency structure places TD-10 on Analytics Warehouse Loader ahead of two other items that depend on it. register item TD-10 has no single accountable owner, and that is the finding the reviewer would press hardest.

Status on Battery Management Service item TD-21 is in progress; the target is 2026-09-30. 1 engineer-quarter is the estimate on programme item TD-21, which converts to $138,000. TD-21 carries this observation: data collection is running at the two affected sites. quality item TD-21 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Safety item TD-19 covers the work to measure closing rate for the separation monitor. TD-19 is in progress with a target date of 2026-08-31. The most significant open safety item in the reviewed record is the point the reviewer would put to management on Speed and Separation Monitor item TD-19. TD-19 is reported here in full because a summary of it in an earlier draft of TDR-2026-34 was misread as an assurance.

The estimate for safety item TD-20 is 1.5 engineer-quarters and $162,000. As at the review date register item TD-20 was in progress, with 2026-09-15 still the stated target. Safety item TD-20 carries this observation: the prototype uses a network-copyleft component that must be replaced before shipping. TD-20 was excluded from an earlier draft of TDR-2026-34 and reinstated after the reviewer obtained the supporting record.

The estimate for TD-06 on Simulation Harness is 1 engineer-quarter and $115,000. Progress on TD-06 on Simulation Harness is recorded as blocked against a 2026-11-30 date. The quality entry at TD-06 is to make the nightly simulation suite blocking. quality item TD-06 is the entry that changed most between the first and second document requests behind TDR-2026-34.

Schedule K - Penetration-test findings by severity with residual risk (TDR-2026-34)

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-01	High	7.4	Identity and Authorization	2025-09-16	2025-11-14	low once lifetimes are cut to one hour with refresh
SEC-2025-02	High	8.1	Release Orchestrator	2025-09-17	2025-09-19	low; redaction is enforced in the logging library
SEC-2025-03	Medium	6.2	Remote Operations Console	2025-09-18	2026-01-22	low; pinning is applied at the gateway and the agent
SEC-2025-04	Medium	5.4	Map Authoring Studio	2025-09-18	2025-11-27	medium; the shortened publication path is still outstanding
SEC-2025-05	High	7.8	Telemetry Ingest	2025-09-19	2026-02-06	low; attestation is now required on the ingest path
SEC-2025-06	Medium	6.5	Video Snippet Service	2025-09-19	2025-12-18	low; the index is scoped per tenant and the identifiers are opaque
SEC-2025-07	Medium	5.9	Site Provisioning Service	2025-09-22	2026-01-16	medium; two forked templates were not re-issued until May 2026
SEC-2025-08	Medium	6.0	Event Bus	2025-09-23	2025-10-31	low; the interface is restricted to the operations subnet
SEC-2025-09	Medium	6.3	OTA Updater	2025-09-24	pending	medium; a two-person rule is designed but not enforced in the tool
SEC-2025-10	Low	3.7	Charge Scheduler	2025-09-24	2025-10-09	low
SEC-2025-11	Low	3.1	Remote Operations Console	2025-09-25	2025-10-30	low
SEC-2025-12	Low	4.2	Analytics Warehouse Loader	2025-09-25	not planned	medium; management accepted the risk with a compensating audit trail
SEC-2026-01	High	7.1	Release Orchestrator	2026-04-14	committed 2026-08-31	high until the second-approver control lands
SEC-2026-02	High	7.6	Robot Shadow Store	2026-04-15	2026-05-29	low; authorization is checked on both the cache and durable paths
SEC-2026-03	High	8.4	Config Distribution	2026-04-15	2026-06-12	low; the flag was removed and the agent refuses unsigned bundles
SEC-2026-04	Medium	5.6	WMS Connector	2026-04-16	2026-05-20	low; payload logging requires an explicit per-tenant approval
SEC-2026-05	Medium	5.2	Alarm Router	2026-04-16	committed 2026-09-30	medium; per-region credentials are scheduled but not issued
SEC-2026-06	Medium	6.1	Simulation Harness	2026-04-17	pending	medium; the role was scoped down but still reads one production bucket
SEC-2026-07	Medium	6.7	Hardware-in-the-Loop Rig	2026-04-17	committed 2026-10-31	medium; network segmentation is the only current

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
						control
SEC-2026-08	Medium	6.4	Identity and Authorization	2026-04-18	2026-06-30	low; role grants now require a second approver and are alerted
SEC-2026-09	Low	3.9	Fleet Health Monitor	2026-04-18	2026-05-08	low
SEC-2026-10	Low	4.0	Billing Usage Meter	2026-04-19	2026-06-05	low; object lock is applied and verified nightly
SEC-2026-11	Low	4.4	Gripper Controller	2026-04-19	committed 2026-11-30	medium; the safety controller limit is the compensating control
SEC-2026-12	Low	3.5	Edge Gateway Agent	2026-04-20	pending	medium; a wipe procedure exists but was not applied retroactively
SEC-2026-13	Low	3.3	Order Broker	2026-04-20	2026-05-15	low
SEC-2026-14	Low	3.8	Traffic Manager	2026-04-21	2026-06-18	low
SEC-2026-15	Low	3.0	Site Provisioning Service	2026-04-21	not planned	low; management judged the operational cost of enforcement higher than the risk
SEC-2026-16	Medium	5.8	Costmap Fusion	2026-04-22	2026-06-26	low; the socket binds to the loopback interface only
SEC-2026-17	High	7.9	Motion Planner	2026-04-22	2026-06-19	low; the port requires a signed local credential
SEC-2026-18	Low	3.4	Battery Management Service	2026-04-23	committed 2026-12-15	low; no fleet impact demonstrated
SEC-2026-19	Medium	5.1	Localization Engine	2026-04-23	committed 2026-10-15	medium; disk encryption is planned for the next compute revision
SEC-2026-20	Low	3.2	Payload Scale Interface	2026-04-24	not planned	low; management accepted pending the calibration service redesign

4. The services carrying the largest debt

Behind Localization Engine sits a local map cache with a periodic checkpoint; the service itself is C++17 with a Rust bridge layer on on-robot Linux, pinned to a dedicated core. Localization Engine is deployed edge, with map distribution from the cloud. Supportability of Localization Engine turns on one fact: an unmaintained scan-matching library is still linked in the fallback path. Autonomy Core releases Localization Engine every three weeks with the planner, and no release in the sampled window bypassed the gate set. On Localization Engine the conclusion is qualified: management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Localization Engine is the reason one section of TDR-2026-34 concludes narrowly rather than generally.

For Localization Engine the reviewer worked from the localization design note, survey reports from three sites and the reviewer's map audit. The interface Localization Engine publishes is versioned, and the motion planner and the fleet health monitor were checked for use of any undocumented field. Inputs to Localization Engine come from the lidar driver, the wheel odometry filter and the map distribution channel, and its consumers are the motion planner and the fleet health monitor. 16 items sit in the Localization Engine backlog; the entry that bears on the investment case is that relocalisation after a lift-and-place takes longer than the operator prompt allows. The published objective for Localization

Engine is that pose error stays under 25 mm in mapped aisles at survey speed. Localization Engine is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Map Authoring Studio is deployed cloud, single global deployment. Applications publishes one objective for Map Authoring Studio: a published map revision reaches every robot at the site inside fifteen minutes. Implementation of Map Authoring Studio rests on TypeScript with a Python service layer; the process runs on cloud, a desktop-class web application and persists to PostgreSQL 14 with map revisions in object storage. The cadence recorded for Map Authoring Studio is weekly, which matches the observed release history. Supportability of Map Authoring Studio turns on one fact: a geometry rendering library one major version behind its supported line. Map Authoring Studio carries a follow-up request that was satisfied after the draft of TDR-2026-34 was circulated.

For Map Authoring Studio the reviewer worked from the authoring workflow document, the revision history for six sites and an export audit. Map Authoring Studio is owned by the Applications team, with Lucia Ferrante named as the engineering owner of record. Interface discipline around Map Authoring Studio is adequate; the schema is versioned and change is gated on a compatibility check. Against Map Authoring Studio, the coverage figure is 49 per cent line coverage, which the number is credible and the measurement basis is honest. Applications tracks 22 open defects on Map Authoring Studio, including the acknowledged case where an unpublished draft can be exported and hand-loaded, bypassing the review step. Map Authoring Studio is reported at the evidence strength available and has not been promoted to fit the narrative.

The backlog for Fleet Health Monitor is 17 items deep and contains an acknowledged defect whereby alert grouping collapses distinct faults at the same site into one case, hiding the second fault. Placement of Fleet Health Monitor is cloud, regional, which the reviewer tested by tracing a request end to end. Fleet Health Monitor shows 57 per cent line coverage in the pipeline, and the number is credible and the measurement basis is honest. Supportability of Fleet Health Monitor turns on one fact: none identified. For Fleet Health Monitor the reviewer worked from the alert rule inventory, three months of case records and a rule coverage review. Fleet Health Monitor is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Fleet Health Monitor has a single named owner, Wanda Njoku of Field Reliability, which is not true of every service reviewed. The telemetry ingest path and the robot shadow store feed Fleet Health Monitor; a failure in Fleet Health Monitor is felt by the alarm router and the field service workflow. Where Fleet Health Monitor is concerned, the commitment is that a fleet-affecting condition raises an alert inside three minutes of first symptom; the reviewer traced the metric to its source. Fleet Health Monitor releases fortnightly. One coupling concern applies to Fleet Health Monitor: a time-series store plus PostgreSQL 14 for case records is shared with a second service, so a schema change is not a local decision. Fleet Health Monitor is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The lidar driver, the depth camera bridge and the wheel odometry filter feed Costmap Fusion; a failure in Costmap Fusion is felt by the motion planner and the speed and separation monitor. Costmap Fusion rests on the fusion design note, sensor timing captures and the reviewer's own bench replay so far as this review is concerned. Costmap Fusion publishes a stable interface, though two fields used by the motion planner and the speed and separation monitor are documented only in code comments. Costmap Fusion pairs a C++17 implementation with on-robot Linux, pinned to the drive control core, and its data lives in a shared-memory ring buffer with no durable persistence. Costmap Fusion was tested against the dependency ageing register, which records a vendored geometry library whose upstream branch left support in 2024. Costmap Fusion is the entry that changed most between the first and second document requests behind TDR-2026-34.

Costmap Fusion releases monthly, batched with perception model updates. Ownership of Costmap Fusion sits with Perception; Daniel Okafor signs off changes to it. Where Costmap Fusion is concerned, the commitment is that fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms; the reviewer traced the metric to its source. Against Costmap Fusion the backlog runs to 21 items, and it includes a known behaviour whereby a stale lidar frame is retained for one extra cycle while a driver reconnects. Costmap Fusion sits edge, co-scheduled with the planner on the compute module, and no alternative placement is currently supported. Costmap Fusion sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

End-of-life review of Gripper Controller found none; the coprocessor part is current production. The Robot Firmware team carries Gripper Controller on its service inventory, and Ines Duarte answered for it in interview. The evidence base for Gripper Controller was the gripper control specification, calibration records for eleven units and a force bench test. The objective recorded against Gripper Controller is that a grasp completes or reports failure inside 1.2 s, and it is reported weekly to the engineering leadership. The topology for Gripper Controller is edge, only on the two robot variants fitted with a manipulator, and the reviewer confirmed it against the deployment manifests. Gripper Controller was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Gripper Controller is promoted quarterly, with the ring order fixed by policy. One coupling concern applies to Gripper Controller: a local parameter store is shared with a second service, so a schema change is not a local decision. Gripper Controller reports 41 per cent line coverage, measured by the same instrumentation the pipeline uses. For Gripper Controller the reviewer records that the reviewer accepts the engineering position while noting that the verifying test does not exist. Gripper Controller depends on the payload scale interface and the perception grasp proposals and is depended on by the order execution path and the safety controller fault input. Gripper Controller is the reason one section of TDR-2026-34 concludes narrowly rather than generally.

Interface discipline around Payload Scale Interface is adequate; the schema is versioned and change is gated on a compatibility check. For Payload Scale Interface the stated target is that payload mass is reported within 2 per cent of the reference load, expressed against production traffic rather than a benchmark. Responsibility for Payload Scale Interface was confirmed with Ines Duarte, whose team, Robot Firmware, also holds the on-call duty. Placement of Payload Scale Interface is edge, only on robots fitted with the weighing deck, which the reviewer tested by tracing a request end to end. Payload Scale Interface ships as needed; the reviewer sampled the last twelve releases. Payload Scale Interface was raised again with the owning engineer after the initial interview, and the position did not change.

11 items sit in the Payload Scale Interface backlog; the entry that bears on the investment case is that the calibration record is stored locally and is lost when the compute module is swapped. A local calibration record is the system of record for Payload Scale Interface, a C++17 component running on on-robot Linux, user-space driver. Documents and data inspected for Payload Scale Interface were the interface specification, calibration records for nine units and a module-swap test. Payload Scale Interface was assessed against the objective and the defect record, and the component is adequate today and carries a supportability liability that is not on the funded plan. Reading the dependency map from Payload Scale Interface outward gives the load cell hardware on the input side and the gripper controller and the order execution path on the output side. Payload Scale Interface was cross-checked against the engineering record supplied for TDR-2026-34, and the two agree.

Behind Fleet Allocator sits PostgreSQL 14 with a synchronous standby; the service itself is Go 1.22 on cloud, containers on a managed Kubernetes cluster. Fleet Allocator was checked for shared-database coupling, and the ownership boundary held. The traffic manager, the charge scheduler and the fleet health monitor rely on Fleet Allocator, which in turn relies on the order broker and the robot shadow store. Taking the evidence together, Fleet Allocator management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Coverage for Fleet Allocator was 82 per cent line coverage at the review date, and the trend over the reviewed period was up by four points. Fleet Allocator has no single accountable owner, and that is the finding the reviewer would press hardest.

The published objective for Fleet Allocator is that task assignment inside 900 ms of order arrival at p95. On paper Fleet Allocator releases weekly, behind a change approval, and the observed practice did not diverge from that. Fleet Allocator runs cloud, one regional deployment serving all sites in that region; that placement is what determines whether a fault is one robot or one region. The Fleet Services team carries Fleet Allocator on its service inventory, and Marta Kellerman answered for it in interview. Fleet Allocator was tested against the dependency ageing register, which records none identified in the reviewed manifests. Fleet Allocator is the reason one section of TDR-2026-34 concludes narrowly rather than generally.

Battery Management Service carries a single measurable objective, namely that pack cell imbalance is reported before it reaches the derating threshold. Battery Management Service was weighed on its coverage, its defect backlog and its dependency position, and the component would not block the transaction, and it would appear on any competent post-close engineering plan. Battery Management Service was built in C on the pack controller, Python 3.11 in the cloud and

has not been rewritten; it still runs on split: pack firmware and a cloud analytics service against pack non-volatile memory and a cloud time-series store. Robot Firmware holds Battery Management Service, and the ownership entry naming Peter Halvorsen matches the on-call rotation. Battery Management Service publishes a stable interface, though two fields used by the charge scheduler and the fleet health monitor are documented only in code comments. Battery Management Service sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Battery Management Service releases pack firmware twice a year, the cloud service monthly. Underlying Battery Management Service in this report are the pack firmware release notes, cell test data for two revisions and a derating incident file. Automated test coverage on Battery Management Service stands at 54 per cent line coverage. Battery Management Service depends on the pack sensors and the charge scheduler and is depended on by the charge scheduler and the fleet health monitor. Battery Management Service's dependency ageing was checked: one pack revision uses a cell supplier that has since discontinued the part. Battery Management Service was checked against the company's own register, which had not been updated since the change landed.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2026-34)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
lint and format	1 m 10 s	1 m 40 s	3.1%	0.1%	blocking	Developer Productivity
unit tests, cloud services	6 m 20 s	9 m 05 s	5.4%	1.9%	blocking	Fleet Services
unit tests, robot software	11 m 45 s	18 m 30 s	7.8%	4.6%	blocking	Autonomy Core
static analysis, robot software	14 m 05 s	22 m 10 s	6.2%	0.4%	blocking	Autonomy Core
build, cloud container images	8 m 15 s	12 m 40 s	2.7%	0.3%	blocking	Platform Infrastructure
build, robot firmware images	19 m 30 s	31 m 20 s	4.1%	0.6%	blocking	Robot Firmware
integration tests, cloud	24 m 50 s	41 m 15 s	12.6%	8.2%	blocking	Fleet Services
contract tests, warehouse connectors	9 m 40 s	15 m 55 s	9.3%	3.7%	blocking	Integrations

4. The services carrying the largest debt - part 2 of 2

Below, the 4. The services carrying the largest debt continues; nothing separates this part from the last except the page on which it happens to fall. The parts are to be read together.

Qualifications noted at the head of the 4. The services carrying the largest debt apply to these rows equally, and are not restated for each part. Pagination alone determines where it falls.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2026-34) (continued, part 2 of 2)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
simulation suite, short	33 m 10 s	52 m 00 s	8.9%	5.1%	blocking	Autonomy Core
simulation suite, nightly	3 h 40 m	4 h 25 m	14.2%	6.8%	advisory	Autonomy Core
hardware-in-the-loop, bench	1 h 15 m	2 h 05 m	10.4%	2.2%	blocking for firmware	Robot Firmware
dependency and licence scan	4 m 25 s	6 m 50 s	1.8%	0.2%	advisory	Security Engineering
secret scan	2 m 05 s	3 m 15 s	0.9%	0.1%	blocking	Security Engineering
container image vulnerability scan	5 m 35 s	8 m 20 s	16.7%	1.1%	advisory	Security Engineering
performance smoke, cloud	12 m 20 s	19 m 45 s	6.6%	3.4%	advisory	Platform Infrastructure
release packaging and signing	7 m 50 s	11 m 10 s	2.2%	0.2%	blocking	Developer Productivity

5. Debt with a capacity consequence

Measurement of single site, steady state run LT-2025-11 is against the internal objective that no telemetry is lost for an outage under four hours rather than against a customer commitment. Production-sized data but a single availability

zone describes the environment single site, steady state run LT-2025-11 ran in. 120-robot run LT-2025-11, taken with the other runs in the series, shows that the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. scenario LT-2025-11 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

Latency in single site, peak hour run LT-2025-12 was 265 ms at p50, 690 ms at p95 and 1,240 ms at p99, with an error rate of 0.05%. The judgement the reviewer draws from single site, peak hour run LT-2025-12 is that the error rate rather than the latency is the term that would first breach a customer commitment. Applying the fifteen-minute map publication objective to scenario LT-2025-12 produces the verdict recorded above. run LT-2025-12 illustrates why TDR-2026-34 separates the severity of a finding from the confidence behind it.

Load run LT-2025-13 was executed in the performance environment, which carries the production cloud configuration and a laboratory network profile. Outcome of load run LT-2025-13 was pass with observation, and the reviewer read the raw result rather than the summary. On single site, peak with a charger outage run LT-2025-13 the reviewer notes that the pass verdict depends on a target that was last recalibrated in 2024. single site, peak with a charger outage run LT-2025-13 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The test bed for run LT-2025-14 carries a synthetic robot fleet driven from recorded mission traces, which is the main reason the reviewer does not treat it as production-equivalent. Run LT-2025-14 was marked pass by the performance team. Load run LT-2025-14 sits against the fifteen-minute map publication objective; whether that target is the right one is a separate question. load run LT-2025-14 would be a candidate for the first post-close engineering review under any reasonable plan.

The environment behind run LT-2025-15 carries the current production topology with one region held out, and the reviewer weighted the result accordingly. Outcome of scenario LT-2025-15 was pass with observation, and the reviewer read the raw result rather than the summary. The four sites, shared region run at four sites, shared region run LT-2025-15 was executed on 2025-10-16 at 480 robots and 9,400 orders an hour. four sites, shared region run LT-2025-15 is reported here in full because a summary of it in an earlier draft of TDR-2026-34 was misread as an assurance.

Scenario LT-2025-16 is recorded as fail against the 1,200 ms p95 target. The point of scenario LT-2025-16 within the series is whether the read replicas were enabled. Scenario LT-2025-16 ran against an environment with a smaller order history than production and a synthetic warehouse endpoint. LT-2025-16 was excluded from an earlier draft of TDR-2026-34 and reinstated after the reviewer obtained the supporting record.

Order broker journal write queue depth exceeded its alert threshold was the constraint order burst, ten times nominal for two minutes run LT-2025-17 exposed. Scenario LT-2025-17 ran against an environment with production data volumes and a stubbed paging provider. Order burst, ten times nominal for two minutes run LT-2025-17 is best read against the neighbouring run in the same series, from which it differs in the warehouse system endpoint used. load run LT-2025-17 has no single accountable owner, and that is the finding the reviewer would press hardest.

On load run LT-2025-18 the reviewer notes that the improvement between the two runs is real and its durability is untested under a shard rebalance. The limiting term in load run LT-2025-18 was reproduced the November 2025 control delay before the topic split. Telemetry burst with control traffic on the shared topic is the scenario at scenario LT-2025-18: 480 robots, 9,400 orders an hour, dated 2025-10-22. LT-2025-18 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Load run LT-2025-19 used the standard performance environment, which is characterised by a replayed order stream taken from the two largest sites. The reviewer's reading of region failover during peak run LT-2025-19 is that the measurement is credible and the extrapolation the company draws from it is not. 480-robot run LT-2025-19 produced a 410 ms median with 1,950 ms and 6,700 ms tails and an 2.85% error rate. run LT-2025-19 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Comparing scenario LT-2025-20 with its neighbour in the series, the difference is whether control and telemetry shared a topic. The result recorded for LT-2025-20 is pass with observation, which the reviewer traced to the underlying measurement. The target database failover, allocator primary run LT-2025-20 was measured against is the internal

objective that no telemetry is lost for an outage under four hours. run LT-2025-20 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Run LT-2026-01 came out pass. Comparing six sites after the topic split run LT-2026-01 with its neighbour in the series, the difference is one platform change between otherwise identical runs. In run LT-2026-01 the bottleneck was the control topic split removed the telemetry coupling. six sites after the topic split run LT-2026-01 illustrates why TDR-2026-34 separates the severity of a finding from the confidence behind it.

The environment behind LT-2026-02 carries a replayed order stream taken from the two largest sites, and the reviewer weighted the result accordingly. 960 robots and 18,200 orders an hour define run LT-2026-02, run on 2026-02-12. LT-2026-02 was marked fail against the 1,200 ms p95 target by the performance team. run LT-2026-02 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Configuration of the environment for 960-robot run LT-2026-03 included a replayed order stream taken from the two largest sites. The result recorded for run LT-2026-03 is pass, which the reviewer traced to the underlying measurement. Weighed against the projected fleet, load run LT-2026-03 indicates that the improvement between the two runs is real and its durability is untested under a shard rebalance. LT-2026-03 carries a follow-up request that was satisfied after the draft of TDR-2026-34 was circulated.

Schedule P - Load-test matrix against the published latency targets (TDR-2026-34)

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-11	single site, steady state	120	1,850	210 ms	480 ms	760 ms	0.02%	pass
LT-2025-12	single site, peak hour	120	2,640	265 ms	690 ms	1,240 ms	0.05%	pass
LT-2025-13	single site, peak with a charger outage	120	2,640	310 ms	870 ms	1,690 ms	0.11%	pass with observation
LT-2025-14	two sites, shared region	240	4,900	290 ms	820 ms	1,510 ms	0.09%	pass
LT-2025-15	four sites, shared region	480	9,400	355 ms	1,120 ms	2,240 ms	0.18%	pass with observation
LT-2025-16	six sites, shared region	720	13,800	470 ms	1,680 ms	3,410 ms	0.34%	fail against the 1,200 ms p95 target
LT-2025-17	order burst, ten times nominal for two minutes	240	12,000	620 ms	2,350 ms	5,900 ms	1.24%	fail against the 1,200 ms p95 target
LT-2025-18	telemetry burst with control traffic on the shared topic	480	9,400	540 ms	4,100 ms	11,300 ms	0.42%	fail
LT-2025-19	region failover during peak	480	9,400	410 ms	1,950 ms	6,700 ms	2.85%	fail against the error rate target
LT-2025-20	database failover, allocator primary	240	4,900	340 ms	1,410 ms	4,200 ms	0.96%	pass with observation
LT-2026-01	six sites after the topic split	720	13,800	395 ms	1,090 ms	2,050 ms	0.14%	pass
LT-2026-02	eight sites, shared region	960	18,200	455 ms	1,340 ms	2,880 ms	0.27%	fail against the 1,200 ms p95 target
LT-2026-03	eight sites with the allocator read replicas enabled	960	18,200	400 ms	1,150 ms	2,310 ms	0.21%	pass
LT-2026-04	ten sites, shared region	1200	22,600	520 ms	1,720 ms	4,050 ms	0.48%	fail against the 1,200 ms p95 target
LT-2026-05	order burst	480	24,000	480 ms	1,240 ms	2,760 ms	0.31%	pass with

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
	after the broker queue change							observation
LT-2026-06	shard rebalance under peak load	720	13,800	430 ms	1,280 ms	3,120 ms	0.36%	pass with observation
LT-2026-07	identity service token storm	960	18,200	180 ms	520 ms	1,010 ms	0.03%	pass
LT-2026-08	console session concurrency	480	9,400	290 ms	740 ms	1,480 ms	0.07%	pass
LT-2026-09	console session concurrency at twice the plan	480	9,400	380 ms	1,460 ms	3,900 ms	0.52%	fail
LT-2026-10	warehouse connector burst, one tenant	240	8,800	610 ms	1,830 ms	4,400 ms	0.68%	fail against the 1,500 ms p95 acknowledgement target
LT-2026-11	warehouse connector burst after outbox partitioning	240	8,800	440 ms	1,210 ms	2,540 ms	0.29%	pass
LT-2026-12	telemetry ingest at twice the current fleet	2400	n/a	1.4 s to console	22 s	48 s	0.08%	pass
LT-2026-13	twelve sites, projected 2027 fleet	1440	27,000	690 ms	2,450 ms	6,100 ms	0.74%	fail
LT-2026-14	four hour edge buffer drain after an outage	720	13,800	not applicable	not applicable	not applicable	0.00%	pass with observation
LT-2026-15	map publication to a thousand robots	1000	n/a	3 m 20 s	8 m 40 s	13 m 10 s	0.10%	pass
LT-2026-16	config distribution with thirty per cent of robots offline	720	13,800	1 m 50 s	4 m 30 s	9 m 20 s	0.00%	pass with observation

6. Where debt produced an incident

Event Bus incident INC-2025-139 touched 7 sites and 246 robots, and multiple sites was the affected account. In capacity incident INC-2025-139, the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited, which is a process finding rather than a code finding. INC-2025-139 took 2 h 47 m to stabilise and 63 days to close out. For capacity incident INC-2025-139 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. The post-mortem for capacity incident INC-2025-139 identifies the cause as follows: control and telemetry traffic shared one topic and one consumer group budget. INC-2025-139 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Capacity incident INC-2025-139 was raised on 2025-11-02 after a telemetry backlog on a shared topic delayed control events by up to eleven seconds. Event Bus incident INC-2025-139 was detected because the traffic manager's own latency alert fired first. Multiple sites incident INC-2025-139 supports one conclusion and not another: the customer-reported detection path is the finding, not the defect itself. For multiple sites incident INC-2025-139 the follow-through was tested rather than assumed, and the split landed on 2026-01-04; the quarterly review has not yet run. The commitment register shows incident INC-2025-139 against an undertaking to split control traffic onto a dedicated topic with its own

quota and revisit the topic inventory quarterly, with Erik Palmgren named and 2026-01-31 committed. Event Bus incident INC-2025-139 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

How resource leak event INC-2025-041 surfaced matters to the assessment: a site supervisor telephoned the on-call engineer after robots queued at one intersection. What Crestline Fulfillment incident INC-2025-041 tells the reviewer is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Resource leak is the class of incident INC-2025-041; Traffic Manager is the component. The trigger recorded for incident INC-2025-041 on 2025-03-14 was that a robot lost its network link mid-traverse and its corridor reservation was never released. In Crestline Fulfillment incident INC-2025-041, the alert rule for queue depth existed but was scoped to a metric the site did not emit, which is a process finding rather than a code finding. resource leak incident INC-2025-041 is one of the entries where TDR-2026-34 relies on the company's own measurement rather than an independent one.

Root cause in Crestline Fulfillment incident INC-2025-041 was that the reservation lease had no expiry, so release depended on the departing robot. The blast radius of Traffic Manager incident INC-2025-041 was 2 sites and 61 robots. Evidence strength on the incident INC-2025-041 analysis is strong on the cause and limited on the contributing factor. The corrective action for Crestline Fulfillment incident INC-2025-041 is to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, and 2025-05-30 is the date the team gave. Mitigation of incident INC-2025-041 took 1 h 52 m and full resolution took 9 days. resource leak event INC-2025-041 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Map Authoring Studio incident INC-2025-131 is recorded as follows on delivery: the export restriction landed on 2025-11-27; the publication work is open. Process bypass incident INC-2025-131 was contained to one site and 34 robots. Process bypass event INC-2025-131 reached the engineering team when a field engineer reported unexpected keep-out zones. 2025-10-14 is the date of Juniper Bay Apparel incident INC-2025-131, on which an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. 3 h 22 m elapsed before process bypass event INC-2025-131 was mitigated, with resolution at 44 days. process bypass event INC-2025-131 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Grading incident INC-2025-131 on evidence rather than on consequence gives moderate on the cause and strong on the delivery position, which was checked in the repository. The corrective action for INC-2025-131 is to restrict export to published revisions and shorten the publication path so the workaround is unnecessary, and 2025-12-15 is the date the team gave. The classification applied to Juniper Bay Apparel incident INC-2025-131 is process bypass, and the affected component is Map Authoring Studio. Contributing to process bypass incident INC-2025-131, and not repaired by the remediation, is that the site had a standing workaround for slow map publication that predated the review step. The engineering team's root cause for incident INC-2025-131 is that the export function did not check publication state, and the reviewer found the analysis sound. incident INC-2025-131 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Internal incident INC-2025-152 would have been shorter or smaller had it not been that bucket creation was not covered by the configuration baseline check. On 2025-12-04 the record for data retention incident INC-2025-152 opens: one region retained operator video clips for ninety days against a thirty-day commitment. As to whether the internal incident INC-2025-152 remediation actually landed: the rule landed on 2026-01-10; deletion of the historical clips is open. The reviewer grades the Video Snippet Service incident INC-2025-152 evidence as strong, since the reviewer reproduced the failure mode in a laboratory environment and has not upgraded the finding to compensate. Taken with the rest of the record, Video Snippet Service incident INC-2025-152 indicates that the blast radius was set by a deployment topology decision made years earlier. Video Snippet Service incident INC-2025-152 carries a follow-up request that was satisfied after the draft of TDR-2026-34 was circulated.

Remediation for incident INC-2025-152 is to apply the lifecycle rule through the configuration baseline and delete the over-retained clips; Anjali Sekhon owns it and the committed date is 2026-01-31. Data retention incident INC-2025-152 shows a short mitigation and a long tail: not applicable; retention only against 37 days. Reported detection applies to incident INC-2025-152, since an internal privacy review found the discrepancy. Cause analysis in data retention incident INC-2025-152 concluded that the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. The affected population in internal incident INC-2025-152 was no production site and no robot directly, the

exposure being to internal systems. data retention event INC-2025-152 carries a follow-up request that was satisfied after the draft of TDR-2026-34 was circulated.

So far as reach is concerned, INC-2025-058 involved 2 site or tenant with no robot taken out of service. The classification applied to data integrity incident INC-2025-058 is data integrity, and the affected component is WMS Connector. Data integrity incident INC-2025-058 is recorded as follows on delivery: landed on 2025-05-30; the backfill was completed on 2025-06-03. The immediate event in data integrity event INC-2025-058 was that two tenants reused the same order identifier namespace and duplicate suppression discarded live orders, dated 2025-05-09. Incident INC-2025-058 produced one commitment: to key suppression on the tenant and identifier pair and backfill the affected order records, owned by Tomas Villareal, dated 2025-07-15. Dunbar Industrial incident INC-2025-058 would be a candidate for the first post-close engineering review under any reasonable plan.

The judgement recorded against INC-2025-058 is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Dunbar Industrial incident INC-2025-058 rests on moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence, which the reviewer states separately from the severity of the finding. What went wrong in WMS Connector incident INC-2025-058 was that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. Data integrity event INC-2025-058 reached the engineering team when the customer reported missing picks the same afternoon. Mitigation of data integrity event INC-2025-058 took 2 h 35 m and full resolution took 21 days. incident INC-2025-058 is the reason one section of TDR-2026-34 concludes narrowly rather than generally.

On 2025-04-21 the record for incident INC-2025-052 opens: an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. Scheduling incident INC-2025-052 supports one conclusion and not another: the incident is unremarkable in itself and revealing about the observability of the surrounding system. The reviewer treats one contributing factor in INC-2025-052 as the more important lesson: the site had changed its shift pattern without notifying the deployment team. Duration for incident INC-2025-052 splits into 4 h 10 m of impact and 16 days to close the underlying cause. Grading INC-2025-052 on evidence rather than on consequence gives limited; the log retention window had closed before the reviewer asked for the raw data. Charge Scheduler incident INC-2025-052 was checked against the company's own register, which had not been updated since the change landed.

Reported detection applies to scheduling event INC-2025-052, since the site's own shift report, escalated the next morning. Landed on 2025-06-11 with a nightly reconciliation report is the position on the scheduling incident INC-2025-052 commitment as at the review date. The affected population in scheduling incident INC-2025-052 was one site and 44 robots. Underneath Evergreen Warehousing incident INC-2025-052 lies the design decision that the scheduler treated a shift calendar entry as immutable once the window had opened. The classification applied to scheduling incident INC-2025-052 is scheduling, and the affected component is Charge Scheduler. scheduling incident INC-2025-052 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

Analytics Warehouse Loader owns data integrity incident INC-2025-110, which the reviewer classes as data integrity. Taken with the rest of the record, Kestrel Medical Supply incident INC-2025-110 indicates that the remediation addresses the immediate cause and leaves the contributing factor in place. Reading Kestrel Medical Supply incident INC-2025-110 from its first entry, on 2025-09-03 a late telemetry partition was excluded from the daily load and a customer report understated throughput. Underneath incident INC-2025-110 lies the design decision that the loader treated a missing partition as empty rather than as incomplete. Duration for INC-2025-110 splits into not applicable; reporting only of impact and 18 days to close the underlying cause. data integrity event INC-2025-110 was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

Detection came through the customer challenged the report three days later, which places incident INC-2025-110 on the reported side of the ledger. The reviewer grades the INC-2025-110 evidence as strong, since the reviewer reproduced the failure mode in a laboratory environment and has not upgraded the finding to compensate. Beyond the direct cause, Analytics Warehouse Loader incident INC-2025-110 shows that the partition completeness check was written but never wired into the load gate. The remediation committed after Analytics Warehouse Loader incident INC-2025-110 was to fail the load on an incomplete partition and reprocess the affected days, owned by Farida Cisse with a due date of 2025-10-31. Landed on 2025-09-21 with a reprocessing run for the affected week is the position on the

Analytics Warehouse Loader incident INC-2025-110 commitment as at the review date. data integrity incident INC-2025-110 was cross-checked against the engineering record supplied for TDR-2026-34, and the two agree.

For INC-2025-071 the reviewer holds strong, resting on the production traces the team supplied evidence and reports the finding at that strength. Detection came through the ingest availability alert fired within ninety seconds, which places INC-2025-071 on the alerted side of the ledger. INC-2025-071 is recorded as follows on delivery: landed on 2025-07-18 and exercised in the September failover drill. Beyond the direct cause, availability event INC-2025-071 shows that the upgrade runbook did not require a partition leadership check between broker restarts. The trigger recorded for availability incident INC-2025-071 on 2025-06-11 was that a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage. availability incident INC-2025-071 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Telemetry Ingest owns multiple sites incident INC-2025-071, which the reviewer classes as availability. Erik Palmgren committed after INC-2025-071 to add a leadership and under-replicated partition check to the broker upgrade runbook, due 2025-07-31. The affected population in INC-2025-071 was 6 sites and 214 robots. What went wrong in multiple sites incident INC-2025-071 was that a broker partition lost its leader during a rolling upgrade of the streaming cluster. Taken with the rest of the record, availability event INC-2025-071 indicates that the incident is unremarkable in itself and revealing about the observability of the surrounding system. multiple sites incident INC-2025-071 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

7. Debt with a security consequence

SEC-2026-15 against Site Provisioning Service stands accepted; not planned is what the record gives for closure. Residual risk after remediation of finding SEC-2026-15 is low; management judged the operational cost of enforcement higher than the risk. For SEC-2026-15 against Site Provisioning Service the record shows that collision created in a staging tenant, which is strong evidence and is graded separately from severity. Finding SEC-2026-15 is rated Low with a numeric band of 3.0 on the tester's scoring scale. assessment finding SEC-2026-15 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Telemetry Ingest finding SEC-2025-05 records that telemetry ingest accepted a robot identity without checking device attestation, discovered on 2025-09-19. As at the review date Telemetry Ingest finding SEC-2025-05 was remediated and the closure entry reads 2026-02-06. Finding SEC-2025-05 reduces to a residual risk of low; attestation is now required on the ingest path. Finding SEC-2025-05 is supported by the fact that the tester submitted synthetic telemetry from an unenrolled host; evidence strength is limited, resting on a configuration read rather than on a demonstrated exploitation. Telemetry Ingest finding SEC-2025-05 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

2026-05-15 is the closure position on finding SEC-2026-13, which the ledger records as remediated. Substantiating SEC-2026-13 against Order Broker is the fact that throughput degradation measured by the tester, graded moderate. On SEC-2026-13 against Order Broker the conclusion is that the fix is verified and the class of defect is not, since no test now prevents its return. SEC-2026-13 records that order broker accepted an oversized payload without a limit, discovered on 2026-04-20. SEC-2026-13 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The reason assessment finding SEC-2026-11 was not downgraded is that a non-privileged process could alter a force limit parameter. On Gripper Controller finding SEC-2026-11 the conclusion is that the underlying pattern appears in two other components that the tester did not reach. 2026-04-19 is the discovery date of finding SEC-2026-11, which finds that gripper controller parameter store writable by any on-robot process. Residual risk after remediation of SEC-2026-11 against Gripper Controller is medium; the safety controller limit is the compensating control. SEC-2026-11 against Gripper Controller is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Identity service admin role assignable without a second approval is the substance of SEC-2026-08 against Identity and Authorization, raised on 2026-04-18. Medium-severity finding SEC-2026-08 leaves residual risk that the reviewer assesses as low; role grants now require a second approver and are alerted. 2026-06-30 is the closure position on SEC-2026-08 against Identity and Authorization, which the ledger records as remediated. Medium-severity finding

SEC-2026-08 was graded Medium, band 6.4, by Meridian Assurance engagement MA-2026-1. SEC-2026-08 is reported at the evidence strength available and has not been promoted to fit the narrative.

Committed 2026-10-31 is the closure position on SEC-2026-07 against Hardware-in-the-Loop Rig, which the ledger records as open. Medium is the rating finding SEC-2026-07 carries, at 6.7 on the assessor's own scale. In Medium-severity finding SEC-2026-07 the exploitation path means an unpatched laboratory host sits on the engineering network. The reviewer's position on SEC-2026-07 is that the compensating control is real but depends on a network boundary that a single misconfiguration removes. Medium-severity finding SEC-2026-07 was inspected in the repository rather than described from the ticket, which is why the account is specific.

Assessment finding SEC-2025-07 is remediated, with a closure date of 2026-01-16. In Medium-severity finding SEC-2025-07 the exploitation path means the same secret was reusable at any site provisioned from that template. Finding SEC-2025-07 produces this judgement: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. Rating for SEC-2025-07 is Medium, band 5.9; the reviewer re-read the scoring rationale and accepted it. SEC-2025-07 against Site Provisioning Service was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

Remediation status on Medium-severity finding SEC-2025-09 is partially remediated; the record shows pending. SEC-2025-09 against OTA Updater was opened on 2025-09-24 and states that oTA artefact signing keys held by three engineers without a quorum control. The reviewer declines to soften SEC-2025-09: the fix is verified and the class of defect is not, since no test now prevents its return. SEC-2025-09 is rated Medium with a numeric band of 6.3 on the tester's scoring scale. SEC-2025-09 against OTA Updater illustrates why TDR-2026-34 separates the severity of a finding from the confidence behind it.

Residual exposure on SEC-2026-18 reads low; no fleet impact demonstrated. Battery pack diagnostic interface has no rate limit is the substance of Low-severity finding SEC-2026-18, raised on 2026-04-23. In SEC-2026-18 the exploitation path means a diagnostic flood can slow pack telemetry reporting. Tracking Battery Management Service finding SEC-2026-18 to closure gives a status of open and a date of committed 2026-12-15. Low-severity finding SEC-2026-18 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Costmap Fusion finding SEC-2026-16 reduces to a residual risk of low; the socket binds to the loopback interface only. At 5.8 the score on SEC-2026-16 against Costmap Fusion places it in the Medium group. Assessment finding SEC-2026-16 stands remediated; 2026-06-26 is what the record gives for closure. 2026-04-22 is the discovery date of Costmap Fusion finding SEC-2026-16, which finds that costmap fusion diagnostic socket bound to all interfaces. SEC-2026-16 against Costmap Fusion was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

The reviewer's position on Traffic Manager finding SEC-2026-14 is that the compensating control is real but depends on a network boundary that a single misconfiguration removes. Finding SEC-2026-14 matters because a manual corridor override cannot be attributed after the fact. Substantiating assessment finding SEC-2026-14 is the fact that action performed and the audit trail checked, graded moderate. Remediated is the status of Low-severity finding SEC-2026-14, dated 2026-06-18. SEC-2026-14 against Traffic Manager is the clearest instance in TDR-2026-34 of a control that exists on paper and is weaker in practice.

The reviewer declines to soften SEC-2026-01: the severity is right and the ownership is unclear, since two teams each believe the other holds it. The finding at SEC-2026-01, dated 2026-04-14, is that emergency release path permits a single approver. Assessment finding SEC-2026-01 is supported by the fact that the tester used the path in a staging tenant and the audit trail recorded no justification; evidence strength is limited, resting on a configuration read rather than on a demonstrated exploitation. One insider can move an unreviewed artefact to production is the exposure SEC-2026-01 describes. SEC-2026-01 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

On Localization Engine finding SEC-2026-19 the tester's evidence was that cache read from a removed module, and the reviewer grades it limited, resting on a configuration read rather than on a demonstrated exploitation. Localization Engine finding SEC-2026-19 was graded Medium, band 5.1, by Meridian Assurance engagement MA-2026-1. Tracking assessment finding SEC-2026-19 to closure gives a status of open and a date of committed 2026-10-15. The reviewer's

position on SEC-2026-19 against Localization Engine is that the severity is right and the ownership is unclear, since two teams each believe the other holds it. SEC-2026-19 against Localization Engine is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

8. Debt with a safety consequence

Reading HZ-09 against the verification evidence, the slip detector is prototyped and unvalidated as at the review date. Field history for hazard HZ-09 is that it has produced no event, though the control has activated repeatedly. Ramp testing on a wet and a dry surface at two sites is what verified Serious injury hazard HZ-09. Remote-exposure hazard HZ-09 is reported here in full because a summary of it in an earlier draft of TDR-2026-34 was misread as an assurance.

On the register's own scale, Occasional-exposure hazard HZ-12 is Minor injury severity, Occasional exposure, Low controllability. Occasional-exposure hazard HZ-12 was tested by witnessed reverse test plus a review of the dock layout at four sites, and the reviewer read the resulting record. Tolerable is the residual position on hazard HZ-12 in the register, whose status is verified. HZ-12 is reported here in full because a summary of it in an earlier draft of TDR-2026-34 was misread as an assurance.

The design measure recorded at Occasional-exposure hazard HZ-22 is local yield behaviour with a conservative speed limit and a timed stop. HZ-22 was verified through network loss injection at one site plus a bench network partition test. The scope of hazard HZ-22 in the register is the newest three sites, where commissioning is incomplete. hazard HZ-22 was cross-checked against the engineering record supplied for TDR-2026-34, and the two agree.

One caveat attaches to HZ-05: nine field units retain a pre-production harness revision and are tracked to retrofit. The hazard recorded at HZ-05 is that drive power is not removed when the emergency stop is actuated. Mitigation for hazard HZ-05 is hardwired dual-channel stop chain independent of the compute module. Improbable-exposure hazard HZ-05 would move from a concern to a verified defect if one further measurement were supplied.

Status open and residual risk Not yet tolerable at the higher approach speed are what the register gives for hazard HZ-18 in the register. The design measure recorded at HZ-18 is protective distance calculation with a stated approach speed assumption. The three risk terms on register entry HZ-18 are severity Serious injury, exposure Occasional and controllability Low. Occasional-exposure hazard HZ-18 is the clearest instance in TDR-2026-34 of engineering practice being better than the documentation suggests.

The three risk terms on HZ-06 are severity Serious injury, exposure Occasional and controllability Low. The hazard recorded at HZ-06 is that robot moves unexpectedly during maintenance. Evidence of verification at hazard HZ-06 consists of procedure walk-through with three service technicians and a lockout inventory. Occasional-exposure hazard HZ-06 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

Produced a single event before the current control was fitted is the incident history behind Equipment damage hazard HZ-07. The verification method for Equipment damage hazard HZ-07 is simulation coverage of boundary cases plus a site survey audit. The hazard recorded at hazard HZ-07 in the register is that robot travels into an unmapped area and contacts fixed plant. HZ-07 is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

One qualification on Equipment damage hazard HZ-08 the reviewer would not remove is that the tie-break landed in October 2025 after two stall events. Equipment damage hazard HZ-08 scores Equipment damage on severity; exposure is Frequent and controllability High. Hazard HZ-08 is the entry for the case that two robots collide at a corridor intersection. Frequent-exposure hazard HZ-08 is the entry that changed most between the first and second document requests behind TDR-2026-34.

The residual entry for hazard HZ-20 in the register reads Tolerable where cleaning compliance is evidenced, partially verified. The hazard recorded at Serious injury hazard HZ-20 is that lidar contamination degrades detection without a fault being raised. Evidence of verification at HZ-20 consists of contamination bench test plus a review of cleaning compliance at three sites. hazard HZ-20 in the register would be a candidate for the first post-close engineering review under any reasonable plan.

Hazard HZ-04 carries residual risk of Tolerable once calibration is gated in the service workflow with a status of verified. The hazard recorded at Remote-exposure hazard HZ-04 is that gripper closes on a hand during a manual intervention. Severity at hazard HZ-04 in the register is Serious injury, with the exposure rated Remote and controllability Low. Serious injury hazard HZ-04 appears twice in the working papers for TDR-2026-34 because the first answer was corrected after a follow-up request.

Hazard HZ-01 in the register applies to every site with a mezzanine or ramp. Witnessed approach testing on six robots plus a review of the on-robot safety log is what verified hazard HZ-01 in the register. Risk parameters for HZ-01 are Serious injury severity, Frequent exposure and Low controllability. register entry HZ-01 is the clearest instance in TDR-2026-34 of a control that exists on paper and is weaker in practice.

The scope of HZ-10 is the newest three sites, where commissioning is incomplete. Hazard HZ-10 in the register is mitigated by contactor sequencing with an arc-suppression design in the charge dock. Hazard HZ-10 produced a single event before the current control was fitted, and the register was not updated to reflect it. register entry HZ-10 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

9. Debt exposure by site

On Lehigh Valley, Pennsylvania the record shows that it is commissioned during the connector rework. The Lehigh Valley, Pennsylvania deployment serves Crestline Fulfillment with 64 robots, live since 2024-02. Lehigh Valley, Pennsylvania carries a follow-up request that was satisfied after the draft of TDR-2026-34 was circulated.

On Bethlehem, Pennsylvania the record shows that it is sparse survey features, which is why relocalisation is slow here. Bethlehem, Pennsylvania came online in 2025-11 and now carries 37 robots for Kestrel Medical Supply. Bethlehem, Pennsylvania was tested rather than accepted, and the test is described in the working papers behind TDR-2026-34.

On Edwardsville, Illinois the record shows that it is the newest Crestline site, still ramping. 63% of headroom remains at Edwardsville, Illinois, whose connector targets vendor B. Edwardsville, Illinois appears twice in the working papers for TDR-2026-34 because the first answer was corrected after a follow-up request.

Remaining capacity at Chino, California is 54%, measured against the site's peak order profile. For Chino, California the material fact is that it is the go-live delayed by the forked provisioning template. Chino, California is included because a reader of TDR-2026-34 should be able to disagree with the inference without repeating the work.

Site ST-08 is Kenosha, Wisconsin: Dunbar Industrial, 41 robots, live from 2024-11. Kenosha, Wisconsin is the second opportunity-charging site, and the reviewer weighted the sample accordingly. Kenosha, Wisconsin was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

Whitestown, Indiana was included in the sample because it is the site with the standing map publication workaround. Headroom at Whitestown, Indiana stands at 45%; vendor C is the integrated warehouse system. Whitestown, Indiana is reported here in full because a summary of it in an earlier draft of TDR-2026-34 was misread as an assurance.

Capacity headroom at Reno, Nevada is 57%, and the warehouse system is vendor D. Site ST-11 is Reno, Nevada: Blue Mesa Components, 29 robots, live from 2025-05. Reno, Nevada was one of two items where TDR-2026-34 could not obtain the underlying data within the engagement.

Groveport, Ohio reports 44% headroom against its integration with vendor A. What distinguishes Groveport, Ohio is that it is opportunity charging only, which is where the pack estimator drift appeared. Groveport, Ohio is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Site ST-09 is Tracy, California: Harborline Systems, 58 robots, live from 2025-01. Headroom at Tracy, California stands at 36%; vendor B is the integrated warehouse system. Tracy, California was inspected in the repository rather than described from the ticket, which is why the account is specific.

The Newnan, Georgia deployment serves Ironwood Grocery Group with 66 robots, live since 2025-07. The reason Newnan, Georgia appears here is that it is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall. Newnan, Georgia is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

For Mesquite, Texas the material fact is that it is in acceptance testing at the review date. Headroom at Mesquite, Texas stands at 71%; vendor C is the integrated warehouse system. Mesquite, Texas is treated as material to the investment case rather than as a routine engineering item.

Vendor B is the warehouse system at Wilmington, Ohio, where headroom is 14%. Crestline Fulfillment operates 112 Helioscope robots at Wilmington, Ohio, commissioned in 2023-08. Wilmington, Ohio was checked against the company's own register, which had not been updated since the change landed.

Site ST-17 is Lockbourne, Ohio: Granite Peak Distribution, 19 robots, live from 2026-04. On Lockbourne, Ohio the record shows that it is commissioned after the template baseline check landed. Lockbourne, Ohio appears twice in the working papers for TDR-2026-34 because the first answer was corrected after a follow-up request.

Shepherdsville, Kentucky reports 41% headroom against its integration with vendor C. Fathom Mobility operates 47 Helioscope robots at Shepherdsville, Kentucky, commissioned in 2025-03. Shepherdsville, Kentucky rests on a single source, and the reviewer has said so rather than implying corroboration.

10. Items the company has chosen not to address

The present report treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. Management's declinatures are reported in the present report at face value; the reviewer's disagreement is stated separately and does not overwrite them. Open items in the report at TDR-2026-34 are stated with what would close each one, so a post-close plan can be costed against them. The reviewer has resisted the temptation to tidy this report: the open items are open because the evidence is not there yet. This report is one input to a diligence process and is not a substitute for the company's own engineering judgement.

Wrenfield Technical Advisory LLP will not treat an open item in the report at TDR-2026-34 as closed on a verbal assurance, and has said so to the engineering leadership. Where report TDR-2026-34 cannot conclude, it says what evidence would let it conclude. TDR-2026-34 separates matters that bear on the closing conditions from matters that belong in the first two quarters after close. The transaction reading of the report at TDR-2026-34 is that the engineering risk is priced in the plan rather than in the purchase agreement. Wrenfield Technical Advisory LLP has costed remediation at the company's own loaded engineering rate rather than at a market rate.

Nothing in this report is presented as verified on the strength of an interview alone. Where the report at TDR-2026-34 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. TDR-2026-34 keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. The reviewer asked in each case whether a declined item had a compensating control, and TDR-2026-34 reports the answer even where it is unsatisfactory. The technical-debt register, end-of-life dependency exposure and remediation follow-through is a wide field, and report TDR-2026-34 is explicit about which parts of it were reached.

Post-close monitoring under the present report would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. R. Ostrowski notes in this report that the company's own reporting is adequate for three of the four recommended indicators. This report was prepared by Wrenfield Technical Advisory LLP for Ardent Peak and covers the technical-debt register, end-of-life dependency exposure and remediation follow-through at Helioscope Robotics, Inc. as at 2026-07-15. Wrenfield Technical Advisory LLP was instructed by Ardent Peak to examine the technical-debt register, end-of-life dependency exposure and remediation follow-through; the resulting report is TDR-2026-34, dated 2026-07-15. The present report contains 8 open items, and a reader should assume that number will move rather than settle.

11. Conclusions on the debt position

On the evidence available by 2026-07-15, report TDR-2026-34 concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. For Ardent Peak's purposes, this report concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. The report at TDR-2026-34 suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. None of the findings in this report would in the reviewer's view justify withholding completion, and two would justify a specific undertaking. Report TDR-2026-34 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

Report TDR-2026-34 recommends that 1 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. Nothing in the present report supports a retention or an escrow; the findings are operational and they are funded. R. Ostrowski tested a sample rather than the population in TDR-2026-34, and the sampling basis is stated in each section. Work on TDR-2026-34 ran over 3 weeks and included 13 interviews and read-only access to the engineering repositories. The technical-debt register, end-of-life dependency exposure and remediation follow-through produced more evidence than this report reproduces, and the working papers hold the remainder.

TDR-2026-34 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. R. Ostrowski would add one qualitative item to the pack proposed in report TDR-2026-34: the count of commitments closed without verifying evidence. Deal implications in this report are modest in aggregate and specific in detail: 1 items for the closing checklist, 8 for the plan. For the transaction, this report identifies 1 matters the reviewer would raise before closing and 8 for the post-close engineering plan. R. Ostrowski has kept the working papers behind this report in a form that a second reviewer could re-perform.

Redis 7 for corridor reservations with a PostgreSQL 14 audit trail is the system of record for Traffic Manager, a Go 1.22 component running on cloud, containers on a managed Kubernetes cluster. The motion planner and the remote operations console rely on Traffic Manager, which in turn relies on the fleet allocator and the site map service. For Traffic Manager the stated target is that corridor deadlock detected and broken inside two seconds, expressed against production traffic rather than a benchmark. On paper Traffic Manager releases weekly, paired with the allocator release, and the observed practice did not diverge from that. The interface Traffic Manager publishes is versioned, and the motion planner and the remote operations console were checked for use of any undocumented field. Traffic Manager was tested rather than accepted, and the test is described in the working papers behind TDR-2026-34.

The open defect backlog for Identity and Authorization holds 6 items, the most consequential being that service-to-service tokens are issued with a twelve-hour lifetime, longer than the design intent. Security Engineering holds Identity and Authorization, and the ownership entry naming Sonia Krall matches the on-call rotation. The assessment of Identity and Authorization is not unqualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Hosting for Identity and Authorization is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with a hardware-backed key store. The customer directory federations and the secret store feed Identity and Authorization; a failure in Identity and Authorization is felt by every cloud service and the remote operations console. Identity and Authorization sits inside the sample rather than being an exception to it, which is the reason it is reported at length.